

Dev Patel

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EDUCATION

International Institute of Information Technology, Hyderabad
B.Tech in Computer Science

Hyderabad, India
July 2024 – Present

RELEVANT COURSEWORK

Computer Science: Data Structures & Algorithms, Operating Systems & Networks, Advanced Algorithms, Computer Systems, Automata Theory, Databases and Applications, Machine Learning, ML Security

Development: Design, Testing & Analysis of Software Systems, Internet of Things

Mathematics: Linear Algebra, Real Analysis, Probability and Statistics, Discrete Structures

EXPERIENCE

S-Learn (Learning Management System) | 
Shankar Foundation

January 2026 – April 2026
Hyderabad, India

- Led a 5-member team to develop a digital learning platform utilizing **Prototyping model** via **iterative client feedback**.
- Redesigned 70+ pages of static educational content into interactive, gamified web modules helping **100+ students**.
- Deployed production-ready releases on **Vercel** and **SPCRC servers (IIIT Hyderabad)** with **role-based access control** for teachers, parents, and admins.

PROJECTS

Splitco (Expense Manager) | 

December 2025 – Present

- Built a 3-tier group expense splitting application using **React**, **Node.js**, and **MongoDB**.
- Reduced settlement complexity to at most n-1 edges using **graph-based debt netting** across n transactions.
- Designed a responsive UI with real-time updates across 4 modules: groups, expenses, balances, and settlements.

ContextOS | 

June 2026 - Present

- Engineered a **Hierarchical Memory System** (working, episodic, semantic, archived) mimicking OS caching to virtualize context windows, reducing average token footprint by **65%** per session.
- Developed a budget-aware **Context Scheduler** with **Semantic Cache** to eliminate **redundant LLM calls**, reducing query latency to **sub-15ms**.
- Shipped an **Adaptive RAG** pipeline with a background **Memory Garbage Collector** (removing 95% of duplicate blocks) and a real-time **Observability Dashboard** tracking latency and token usage.

Agri-Bhasha Toolkit | Megathon 2025 | 

October 2025

- Collaborated with a 5-person team to engineer a multilingual agricultural assistant using **Bhashini API** with **OCR-based extraction** for **22 Indian languages**.
- Shipped a 3-service deployment stack across **Vercel** and **Render** managing **high-quality pull requests** and **peer code reviews**.

Docs++ (Distributed Document Editor) | 

October 2025 – November 2025

- Built a collaborative document editor supporting **10+ concurrent users** with **sentence-level locking** over **TCP sockets**.
- Reduced metadata lookup latency to **sub-1ms** using **LRU caching** and **Trie-based indexing**.

UniGate (Campus Security Management) | 

January 2026

- Coordinated with campus authorities to develop a **QR-based campus gate system** with **3-role workflows** (students, guards, admins) and leave tracking via IIIT CAS SSO.
- Integrated **Supabase DB** for live approvals and **WhatsApp Web.js** for automated delivery and overstay alerts.

Smart Glove | 

February 2025 – April 2025

- Designed an IoT-based wearable smart glove using **IMU**, **Flex sensor**, and **ESP32** to support **5+ gesture-driven commands**.
- Evaluated gesture-to-command mappings through iterative testing achieving **100% accuracy**, providing accessibility for Smart Home/Device control.

C-Shell (Unix Shell) | 

August 2025 – September 2025

- Developed a Unix-like shell with AST-based parsing, **pipelines**, and **job control**.
- Added 3 core controls: I/O redirection, background jobs, and signal handling for process management.

xv6 Demand Paging |

September 2025 – October 2025

- Implemented **demand paging** in xv6 with **lazy allocation**, page-fault handling, and on-demand loading of text and data segments.
- Added **FIFO page replacement**, swap-in/swap-out support, and a **memstat** syscall for runtime memory debugging.

TECHNICAL SKILLS

Languages: C, C++, x86-64 Assembly, Python, HTML/CSS, JavaScript, TypeScript, SQL, Bash

Frameworks: React, Vite, Flask, Node.js, Express.js

Tools & Platforms: Git, GitHub, Wireshark, Docker, Postman, LaTeX, Figma, MongoDB, MySQL

Libraries: NumPy, Pandas, Matplotlib

ACHIEVEMENTS

- JEE Mains 2024 - AIR 1139 (Top 0.1%) | JEE Advanced 2024 - AIR 4199 (Top 2%).
- Top 20 Finalist, Megathon 2025 (Agri-Bhasha Toolkit) – IIIT H's largest hackathon with 1000+ participants.
- Marketing and Social Media Head, The Gaming Club – largest club at IIIT Hyderabad with 1200+ followers.
- Coordinated cultural activities for interhouse competitions as a captain of Prithvi House, Cultural Council.